

Research on a New Bidirectional DC-DC Topology for Space Applications

Huo Qunhai, Yin Jingyuan*, Wu Lixin, Wei Tongzhen

The Institute of Electrical Engineering Chinese Academy of Sciences
Beijing China

huoqunhai@mail.iee.ac.cn, yinjingyuan@mail.iee.ac.cn, wulixin@mail.iee.ac.cn, tzwei@mail.iee.ac.cn

Abstract—In the spacecraft power supply system, due to the discontinuous working current, the traditional boost circuit has unstable operating points and leads to the depression of bus power quality and the module efficiency, which can't be directly applied to the space power system. With the advantages of high power density, high efficiency and continuous current, the Weinberg topology has been supplied in the practical space power system. In this paper, we improve the Weinberg topology, which combines the discharge and charge function in one converter, to achieve smaller volume, lighter weight and higher power density space power system. A detailed analysis is made on the bidirectional working principle of the proposed circuit topology, and the circuit gain of boost and buck processing is derived. The simulation model is built to show the feasibility of the proposed topology.

Keywords—Weinberg; Space power; Bidirectional dc-dc; Gain calculation; Working mode

I. INTRODUCTION

With the development of space technology, the demand of the spacecraft power system becomes more and more large [1]. Because the launch cost of the spacecraft is expensive, and the power supply system occupies a large proportion in the total quality of the spacecraft, reducing quality of small power supply system plays a key role in decreasing the satellite launch cost [2]. Furthermore improving the spacecraft power density has a high practical value [3], [4]. The existing research results show that the bidirectional converter has more advantages in the aspects of volume, power density, weight, and so on [5],[6], so the research of bidirectional DC converter becomes more meaningful.

The selected target of the appropriate high power density topology should be simple stable and reliable [7]. Due to the discontinuous working current and unstable operation, the traditional boost circuit can't be directly applied to the space power system. A variety of new topologies have become the research focus on the space power supply, which have been mainly used in aerospace applications: Weinberg [8]-[13], super boost, bidirectional coupled boost [14].

Since the Weinberg topology has been proposed, the isolated and the non-isolated Weinberg topology have been the hot spot [6], [8]-[13]. In recent years, with the rapid increase of power level demand in space power, the volume and weight of passive devices have become an important factor in the development of space power supply.

In order to overcome the shortcomings of the existing topology, this paper proposes a non-isolated bidirectional chopper topology. Then the theoretical analysis and simulation study is completed. The proposed topology has the advantages of simple topology, high power density, charging function and discharging function, which could implement the bidirectional power transmission.

II. WORKING PRINCIPLE OF SPACE POWER SUPPLY

The space power system makes solar energy into electric energy by solar panels convert. But when the earth is located between the satellite and the sun, the satellite cannot receive the sun light to convert solar energy into electrical energy, and the satellite will work in abnormal state. Furthermore, the satellite configures a certain battery and solar arrays for the combined power supply. When the power of the solar energy has the remaining part besides satisfying the power supply of the satellite, the more energy could be stored in the battery. When the satellite is located in a shaded area, the battery could release electric energy to power satellites. So the satellite could work in a stable state. The working principle of the system is shown in Figure 1.

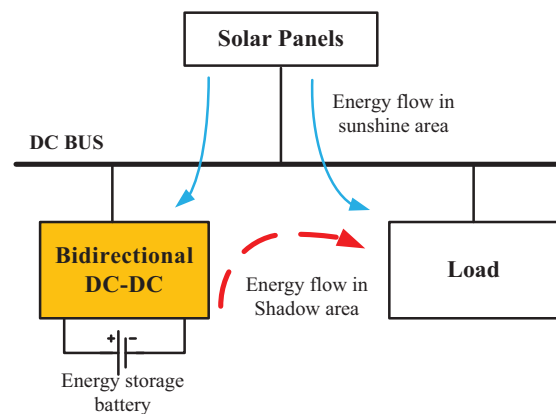


Fig.1 System working principle

When the satellites revolve around the earth, most of the time is being in the sunshine area, so for the charging regulator, only a few modules and low current charging could complete the battery charging. When the satellite works in the shadow area, the total energy demand should be supplied by the discharging regulator which needs multiple high current discharge modules working at the same time. Therefore, the discharge regulator quality of satellite power system plays a key role on the satellite power system cost.

III. PROPOSED TOPOLOGY STRUCTURE

In this paper, based on the conventional Weinberg discharge topology, the proposed topology adds the switching device and the diode to realize the energy bidirectional flow function. The proposed bidirectional chopper circuit topology is shown in Figure 2, which has two working modes: boost mode and buck mode. As seen from Figure 2, the boost part is Weinberg topology, and the buck part is conventional buck circuit. The combination of the two can achieve the bidirectional energy flow function. Because two modes are grouped in one topology, the volume is smaller, and the energy density is higher.

A. Boost mode

The storage battery discharges through the chopper circuit when the spacecraft works in the shadow area, which works in boost mode, to achieve load power supply by the energy storage battery. Through the alternate control of the switch K1 and K2, the storage battery transfers the power to the DC bus, while the switch K3 does not work.

B. Buck mode

The storage battery charges through the chopper circuit when the spacecraft works in the sunshine area, which works in buck mode, realizing the storage battery charging from the solar cell panel. Through the control of switch K3 to achieve power transmission from the solar panels to the power storage battery, while the switch K1 and K2 does not work.

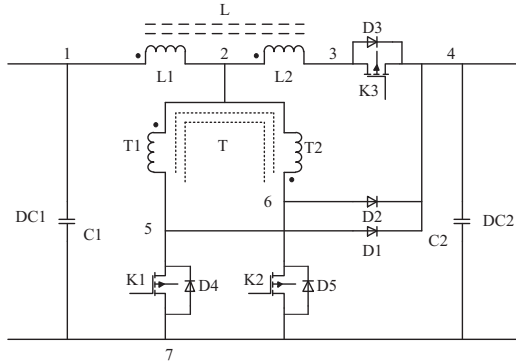


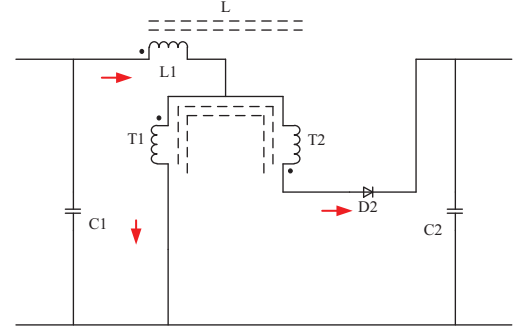
Fig.2 proposed bidirectional chopper circuit topology

IV. WORKING PRINCIPLE

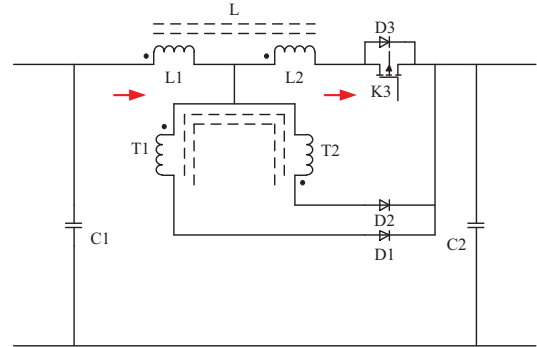
The process analysis of the boost operation principle for the bidirectional DC chopping circuit is shown in Figure 3 (a) and 3 (b). The switch device K1 and the switch device K2 have two times to turn off in one switch cycle. In this way, the switching frequency is effectively increased, and the bandwidth of the converter is rose without increasing the switching loss, meanwhile the size of the magnetic core component and the filter element is reduced. Firstly, assume the switch device K1 is conduction mode when analyses working time sequence, and energy storage inductive is charging by power storage system which discharge load separately. The energy flow diagram is shown in Figure 3 (a). When the switch device K1 is turned off, the load is charged by the power storage system and storage inductor while the dead time of the K1 and K2. The energy flow diagram is shown in Figure 3 (b). Secondly, when the switch device K2 is turned on, the left power storage system charges on inductive

and charging power storage system discharges on right load separately. Lastly, turning off the switch device K2, power system and energy storage inductor discharge on the load while the dead time of K1 and K2. Then the cycle repeats itself that K1 is turned on.

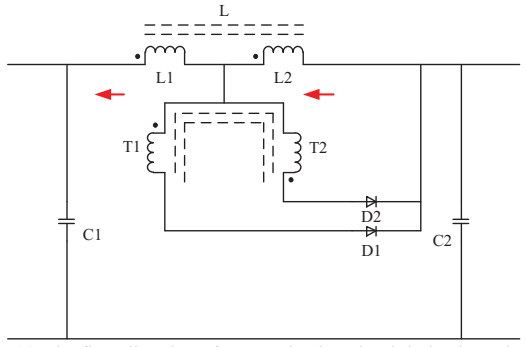
The working process of bidirectional buck DC chopper circuit is shown in Figure 3 (c) and 3 (d). When the switch device K3 is turned on, the storage battery is charged by the right side of the solar array and inductor L. The energy flow diagram is shown in Figure 3 (c). When the switch device K3 is turned off, the battery is charged by inductor L through the diode D4. The energy flow diagram is shown in Figure 3 (d), and the switch device K3 is turned on starting next cycle process.



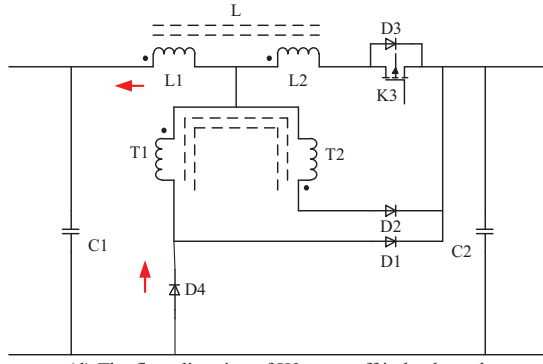
(a) The flow direction of K1 conduction circuit in boost mode



(b) The flow direction of K1 turns off in boost mode



(c) The flow direction of K3 conduction circuit in buck mode



(d) The flow direction of K3 turns off in buck mode
Fig.3 Boost and buck mode working process

V. CIRCUIT GAIN CALCULATION

A. Boost circuit voltage gain

The current of proposed topology flows through the different circuits in the boost and buck process, so the boost and buck voltage gains are different, which need to be calculated separately. First the boost voltage gain is calculated. The capacitor C1 voltage is the input voltage V_{in1} , the capacitor C2 voltage is output voltage V_{out1} , transformer T ratio of T1:T2 is n , and the coupling inductance L1 and L2 coils are N_{L1} and N_{L2} . K1 conduction mode is shown in Figure 3 (a). The relationship of connection point 2 potential V_2 and the output voltage V_{out1} is:

$$V_2 = V_{out1} - nV_2 \quad (1)$$

Then

$$V_2 = \frac{V_{out1}}{n+1} \quad (2)$$

The voltage of Inductance L1 is:

$$V_{L1} = V_{in1} - \frac{V_{out1}}{n+1} = L_{on} \frac{\Delta i_{L1}}{\Delta t} \quad (3)$$

L_{on} is the main inductor of the circuit, then:

$$L_{on} = L_1 = (N_{L1})^2 A_L \quad (4)$$

The coefficient of the coupling inductance is defined as A_L . It is concluded that the current change rate of the inductor L1 in the K1 conduction state is:

$$\frac{\Delta i_{L1}}{\Delta t} = \frac{V_{in1} - \frac{V_{out1}}{n+1}}{L_{on}} \quad (5)$$

Defining the inflow capacitor C2 current as the output current i_{out1} , the relationship between the inductor L1 current and the output current is:

$$i_{L1} = i_{T1} + i_{T2} = ni_{out1} + i_{out1} = (1+n)i_{out1} \quad (6)$$

The rate of current change of output current is:

$$\frac{\Delta i_{out1}}{\Delta t} = \frac{1}{n+1} \frac{\Delta i_{L1}}{\Delta t} = \frac{(n+1)V_{in1} - V_{out1}}{(n+1)^2 L_{on}} \quad (7)$$

When K1 and K2 are completely turned off, the current is shown in Figure 3 (b). The coupling inductor L1 and L2 are all connected in the main circuit, and then at the same time the inductor L_{off} is:

$$L_{off} = L_1 + L_2 = (N_{L1} + N_{L2})^2 A_L = \left(\frac{N_{L1} + N_{L2}}{N_{L1}}\right)^2 L_{on} \quad (8)$$

The voltage between the inductor L1 and L2 are V_{in1} and V_{out1} , and the variation rate of the output current i_{out1} is obtained:

$$\frac{\Delta i_{out1}}{\Delta t} = \frac{V_{in1} - V_{out1}}{L_{off}} = \frac{V_{in1} - V_{out1}}{\left(\frac{N_{L1} + N_{L2}}{N_{L1}}\right)^2 L_{on}} \quad (9)$$

In order to ensure the output current continuously, the current rising volume in the formula (7) is equal to the current dropping volume in the formula (9). The duty cycle of the converter is D_1 , the switching period is T_s , so the rise time of the current in the switching period is $D_1 T_s$, Down time is $(1-D_1)T_s$. Thus the formula (7) and (9) show:

$$\frac{(n+1)V_{in1} - V_{out1}}{(n+1)^2 L_{on}} D_1 = -\frac{V_{in1} - V_{out1}}{\left(\frac{N_{L1} + N_{L2}}{N_{L1}}\right)^2 L_{on}} (1-D_1) \quad (10)$$

Then:

$$V_{out1} = \left[1 + \frac{mnD_1}{mD_1 + (1-D_1)(1+n)^2}\right] V_{in1} \quad (11)$$

Where: $m = \left(\frac{N_{L1} + N_{L2}}{N_{L1}}\right)^2$

When the transformer ratio is 1, and the coupling inductance is the same, the formula (11) can be simplified to:

$$V_{out1} = [1 + D_1] V_{in1} \quad (12)$$

The output voltage is between one and two times of the input voltage.

B. Buck circuit voltage gain

Figure 3 (c) and figure 3 (d) are given a buck process equivalent circuit. The capacitor C1 voltage is defined as the output voltage V_{out2} , capacitor C2 voltage as the input voltage V_{in2} and the inductor L1 current is defined as the output current i_{out2} . The buck process for Figure 3 (c) is the same as the boost process for figure 3 (b). At this time the formula (9) can be drawn the output current change rate:

$$\frac{\Delta i_{out2}}{\Delta t} = \frac{V_{in2} - V_{out2}}{L_{off}} = \frac{V_{in2} - V_{out2}}{\left(\frac{N_{L1} + N_{L2}}{N_{L1}}\right)^2 L_{on}} \quad (13)$$

Figure 3 (d) shows the buck process when the K3 is turned on, and the output voltage acts on the inductor L1. Ignoring the leakage inductance of the transformer, the output current change rate is:

$$\frac{\Delta i_{out2}}{\Delta t} = \frac{V_{out2}}{L_{on}} \quad (14)$$

The K3 conduction (or D4 current inosulation) duty cycle is defined as D2. To ensure the output current is continuous, it is obtained:

$$\frac{V_{in2} - V_{out2}}{\left(\frac{N_{L1} + N_{L2}}{N_{L1}}\right)^2 L_{on}} (1 - D_2) = -\frac{V_{out2}}{L_{on}} D_2 \quad (15)$$

Then

$$V_{out2} = \frac{1}{1 + \frac{mD_2}{1 - D_2}} V_{in2} \quad (16)$$

When the number of the coupling inductor is the same, the voltage gain of the buck circuit is:

$$V_{out2} = \frac{1}{1 + \frac{4D_2}{1 - D_2}} V_{in2} \quad (17)$$

VI. SIMULATION AANALYSIS

It should be noted that the DC chopping circuit is divided into the two work modes which work independently. In the sunshine area, only charge buck mode works, and in the shadow area, only the discharge boost mode works.

In order to verify the feasibility of the proposed topology, a simulation model of boost and buck circuit is built in the PSIM simulation software. The parameters of the simulation system are shown in table 1.

TABLE I. SIMULATION PARAMETERS

Energy storage inductance L	0.144mH
Leakage inductance of transformer T	0.393mH
CapacitanceC1	47uF
CapacitanceC2	47uF
switching frequency	50kHz
DC Bus Voltage	100V
Energy storage terminal voltage	58V

The simulation waveform of boost process is shown in Figure 4, red curve V_{in} is the power supply voltage of the power system, blue curve V_{out} is the output voltage of double loop control, the horizontal axis is time of seconds (s), the

vertical axis is the voltage of volts (V). The starting voltage of output capacitor is zero. After a short period of adjustment for boost control, the load voltage is charged stably at 94V as seen from the simulation waveform. The power transmission is achieved from energy storage system to load, which shows that the topology and control method proposed is effective and feasible.

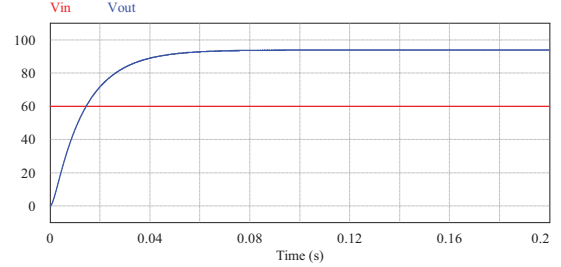


Fig.4 Boost simulation waveform

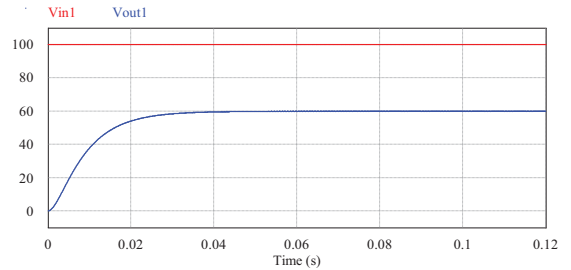


Fig.5 Buck simulation waveform

Buck simulation waveform is shown in Figure 5. The red curve V_{in1} is the charging DC bus voltage of PV, and the blue curve V_{out1} is the output storage battery voltage of double loop control. The horizontal axis is time in seconds (s), and the vertical axis is the voltage in volts (V). The simulation waveform shows storage voltage stability by adjusting a short time, realizing the power transmission from solar system to energy storage system, which shows that the proposed topology is effective and feasible.

VII. CONCLUSION

The bidirectional converter has more advantages in the aspect of volume, power density, and weight and so on. In this paper, based on the non-isolated Weinberg circuit, a bidirectional DC chopping circuit topology is proposed, and the working mode of the boost and buck is analyzed. Simulation model is built based on computer simulation software. The simulation results show that the proposed topology is correct and feasible.

REFERENCES

- [1] DU Qing;CUI Bo;ZENG Yi, et al. Research of space power system based on S3MPR topology and incremental conductance algorithm[J]. Spacecraft Engineering, 2016, 25(3):74-79.
- [2] Ashish Shrivastav; Shikhar Singh; Anirudha Mahajan; Subhashish Bhattacharya. Effective control & software techniques for high efficiency GaN FET based flexible electrical power system for cube-satellites[C]. 2016 IEEE Applied Power Electronics Conference and Exposition (APEC), Year: 2016,Pages: 601 – 608.

- [3] Yawei Liu; Xiaobao Su. High power and efficiency power combining of space TWTAs with waveguide magic-T for Satellite communication[C]. Vacuum Electronics Conference (IVEC), 2015 IEEE International, Year: 2015, Pages: 1 – 2.
- [4] A. Garrigos, J.A. Carrasco, and J.M. Blanes. A Power Conditioning Unit for High Power GEO Satellites Based on the Sequential Switching Shunt Series Regulator[C]. IEEE Mediterranean Electrotechnical Conference, 2006 :1186-1189.
- [5] Ding Lei. Research on magnetcs integrated interleaved DC-DC power module [D]. Harbin Institute of Technology, 2015.
- [6] S. H. Weinberg, A. Lopez. A Bidirectional BDR/BCR for Satellite Applications[C]. Proceedings of the Fifth European Space Power Conference(ESPC). 1998:16-27.
- [7] Zhang Wei. Study of high power destiny DC-DC power supply for space application[D]. Zhe Jiang University, 2009.
- [8] Larico, HRE, Barbi. Three-Phase Weinberg Isolated DC-DC Converter Analysis, Design, and Experimentation[J]. Industrial Electronics, IEEE Transactions on, 2012, 59(2): 888-896.
- [9] Weinberg, A K, Rueda Boldo. A high power, high frequency, DC to DC converter for space applications[C]. Power Electronics Specialists Conference, IEEE Transactions on, 1992, vol.2: 1140-1147.
- [10] S Hawasly, Z Moussaoui, P Kornetzky, et al. Dynamic modeling of parallel connected DC to DC converters using Weinberg topologies[C]. Southcon/96. Conference Record, 1996: 599-609.
- [11] Maset, E Ferreres, A Ejea, et al. 5kW Weinberg Converter for Battery Discharging in High-Power Communication Satellites[C]. 2005 IEEE 36th Power Electronics Specialists Conference, 2005, Pages: 69 – 75.
- [12] Ming Fu; Donglai Zhang; Tiecai Li. Design and Analysis of a Conductance Compensator for Keeping Constant Bandwidth and Output Impedance in Average Current Mode Control[J]. IEEE Transactions on Power Electronics, Year: 2016, Volume: PP, Issue: 99, Pages: 1 – 1.
- [13] Hugo R. E. Larico; Ivo Barbi. Three-Phase Weinberg Isolated DC–DC Converter: Analysis, Design, and Experimentation[J]. IEEE Transactions on Industrial Electronics. Year: 2012, Volume: 59, Issue: 2, Pages: 888 - 896
- [14] B. L. Narasimharaju, S. P. Dubey, S. P. Singh. Coupled inductor bidirectional DC-DC converter for improved performance[C]. Industrial Electronics, Control & Robotics (IECR), 2010 International Conference on 2010: 28-33.